

Qing Lan

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Ph.D. candidate at Peking University and a member of the Beijing International Center for Mathematical Research (BICMR), supervised by Prof. Yi Liu. Expected to graduate in summer 2027; currently seeking postdoctoral positions.

Research Interests

I work in geometric topology, with interests in 3-manifolds, knot theory and hyperbolic geometry. My current work concerns representations of fundamental groups of 3-manifolds and their invariants, carried out with the help of computational tools such as Mathematica, SnapPy, Regina, SageMath, Magma and GAP. I also have broader interests in other aspects of low-dimensional topology and in algebraic methods.

Papers and Preprints

[2] (with Hong Chang and Thiago de Paiva) Twisted torus links that are unlinks. Preprint.

[1] A generalization of Seifert geometry based on the Siegel upper half-space. *Topology and its Applications* **389** (2026), 109870.

Education

- Ph.D. candidate in Mathematics, Peking University (Sep 2022 – present). Advisor: Prof. Yi Liu.
- B.S., Tsinghua University (Sep 2018 – Jul 2022).

Teaching Experience

Teaching assistant at Peking University:

- Spring 2026: Algebra II (Honors Class)
- Fall 2025: Geometry I (Honors Class)
- Spring 2025: Advanced Algebra II
- Fall 2024: Advanced Algebra I
- Spring 2024: Differentiable Manifolds
- Fall 2023: Homology Theory
- Spring 2023: Differential Topology
- Fall 2022: Differentiable Manifolds and Topology (Honors Class)

Talks

- *A Generalization of Seifert Geometry Based on the Siegel Upper Half-Space*. South China University of Technology (Jan 2025); Peking University (Mar 2025); YMSC, Tsinghua University (Apr 2025).

- *Numerical Computation of Seifert Volume of Representations*. Peking University (Jun 2024).
- *Symmetry Break Behavior in Energy Maximizing Graphs*. Peking University (Mar 2024).
- *Turaev–Viro TQFT and the Rank versus Genus Conjecture*. Academy of Mathematics and Systems Science, CAS (Apr 2022).
- *Proofs of Mostow Rigidity Theorem*. Tsinghua University (Oct 2021, Nov 2021); South China University of Technology (Feb 2023).

Personal Website

Further information, including lecture recordings and detailed course syllabi, is available on my personal website.

<https://lanqing-math.github.io/>